

Hindi–Telugu and Telugu–Hindi Machine Translation using a Hybrid Traditional NLP Pipeline

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Abstract—Machine translation (MT) between Hindi and Telugu is a critical task for enabling seamless communication across two of India’s most widely spoken and culturally significant languages. Despite the linguistic richness inherent in both, their structural differences—stemming from their origins in separate language families (Indo-Aryan vs. Dravidian)—make direct translation a computationally and linguistically challenging endeavor. While modern neural network architectures, such as Transformers, currently dominate the global Natural Language Processing (NLP) landscape, they inherently require massive computational resources (GPUs) and extensive parallel datasets. Such resources are often unavailable in low-resource scenarios common to many Indian languages.

This project addresses this gap by developing a traditional, approach-based bilingual translation system capable of converting Hindi to Telugu and Telugu to Hindi. The system leverages a hybrid pipeline that combines rule-based processing, statistical linguistic patterns, and structured data transformations. Instead of relying on “black-box” neural architectures, the system is constructed using interpretable NLP techniques, including Term Frequency-Inverse Document Frequency (TF-IDF) feature extraction, Cartesian product-based bilingual dictionary induction, and 5-gram Maximum Likelihood Estimation (MLE) language modeling. The experimental results demonstrate that such a system provides dependable, explainable translations for short, conversational sentences, offering a viable and efficient alternative for offline and resource-constrained deployment environments.

Index Terms—Machine Translation, Hindi, Telugu, Statistical NLP, TF-IDF, Language Modeling, Low-Resource Computing, Transliteration.

I. Introduction

A. Project Overview

The linguistic landscape of India is one of the most diverse in the world. Machine translation (MT) between Hindi and Telugu plays a crucial role in enabling seamless communication across this diversity. Hindi, an Indo-Aryan language derived essentially from Sanskrit, is predominantly spoken in Northern India. In contrast, Telugu is a Dravidian language with a highly agglutinative structure, spoken predominantly

in Southern India. These two languages represent distinct linguistic typologies.

Despite the cultural exchange between these regions, the structural differences—such as script (Devanagari vs. Telugu script), morphology (inflectional vs. agglutinative), and syntax—make direct translation challenging. Traditional NLP techniques continue to provide a strong foundation for handling such low-resource and morphologically complex languages. These methods are particularly well-suited for scenarios where advanced neural models require large datasets that may not always be available. This project focuses on building a traditional approach-based bilingual translation system capable of converting Hindi to Telugu and Telugu to Hindi by leveraging rule-based processing, linguistic patterns, and structured data transformations.

B. Motivation

The increasing digital presence of Indian languages has created a strong demand for accessible translation systems that support regional communication, education, and information sharing. Many government portals, academic content, and public resources are still available only in one language (often English or Hindi), limiting inclusivity for speakers of other Indian languages like Telugu.

A system that bridges Hindi and Telugu helps achieve several critical societal goals:

- **Migration Support:** It reduces language barriers for students, migrants, and professionals moving between North and South India.
- **Digital Preservation:** It preserves linguistic diversity by making digital content available in multiple regional languages.
- **Research Enablement:** It supports low-resource language research, since Telugu in particular lacks large, high-quality parallel corpora compared to European languages.

- **Resource Constraints:** It enables offline and resource-constrained usage. In many rural Indian contexts, high-end GPUs required for running Transformer models are not available. Traditional methods perform reliably on standard CPUs.

This motivated the development of a structured, traditional machine-translation pipeline that emphasizes linguistic rules, clean preprocessing, and high-quality sentence alignment to provide dependable translations even in low-resource settings.

C. Linguistic Divergence Challenges

One of the primary motivations for a rule-assisted statistical approach over a pure neural approach is the specific nature of the Hindi-Telugu divergence. Understanding these differences is crucial for appreciating the complexity of the translation task.

1) *Morphological Differences:* Hindi relies heavily on postpositions and separate auxiliary verbs to indicate tense and case. It is largely a fusional language where grammatical markers are distinct words. For example, the Hindi phrase "in the house" might be "ghar mein" (two words).

Telugu, conversely, is highly agglutinative. It fuses these markers into a single word stem. The same phrase "in the house" translates to "intlo" (one word, derived from illu + lo). A statistical word-alignment model must handle this many-to-one mapping (Hindi-to-Telugu) or one-to-many mapping (Telugu-to-Hindi). This discrepancy often confuses simple word-based alignment models (like IBM Model 1) if not handled via morphological stemming or robust dictionary lookups.

2) *Syntactic Structure and Sandhi:* While both languages generally follow a Subject-Object-Verb (SOV) order, the internal structure of noun phrases differs. Adjectives and genitive markers may require reordering. Furthermore, Telugu exhibits *Sandhi* (euphonic combination), where the boundaries between words blur due to phonetic fusion at the junction. For instance, two words ending and starting with vowels might merge, altering their spelling. This phenomenon is less prevalent in modern Hindi. A traditional pipeline allows for specific transliteration and dictionary interventions to handle these mismatches that a small-data Neural Machine Translation (NMT) model might miss or "hallucinate."

3) *Script and Phonetics:* Hindi is written in Devanagari, while Telugu has its own script. However, both share a common phonetic ancestry rooted in the Brahmic scripts. They share a similar arrangement of vowels and consonants (Varnamala). This phonetic overlap is the cornerstone of our "Transliteration Fallback" strategy. Even if a word is

unknown, transliterating it preserves the sound, which is often intelligible to the user due to the high volume of Sanskrit loanwords (Tatsama words) present in both languages.

II. State of the Art Work

Machine Translation (MT) for Indic languages has progressed from early rule-based approaches to statistical and neural architectures. This section summarizes key works relevant to traditional MT and Indic language processing.

A. Foundational Corpora and Alignment

The quality of any statistical machine translation system is heavily dependent on the quality of its alignment. Tiedemann (2007) introduced subtitle-based multilingual corpus construction and proposed time-overlap alignment methods effective for noisy, conversational subtitle data [1]. This work influenced later corpus-building pipelines used for Indic languages. The robustness of subtitle alignment is particularly relevant when dealing with the informal and dialogue-heavy nature of available Hindi-Telugu pairs. Subtitles often contain short, colloquial sentences which map well to the capabilities of statistical dictionary induction.

B. Large-Scale Indic Parallel Corpora

The availability of data is the primary bottleneck for Indic MT. The IIT Bombay English-Hindi Parallel Corpus consolidated 1.49M parallel sentence pairs from diverse domains and became an important benchmark for Hindi MT [2]. The authors also reported baseline Statistical Machine Translation (SMT) and NMT results and described robust normalization and tokenization strategies. While this resource is significant for Hindi, similar large-scale parallel corpora for the Hindi-Telugu pair remain scarce, necessitating the use of custom dataset creation methods.

C. SMT vs. NMT for Indian Languages

Das et al. (2023) developed SMT systems for 15 Indian languages, demonstrating that SMT often outperforms NMT in low-resource settings [3]. Their analysis of datasets like Samanantar, OPUS, and FLORES-200 aligns closely with the requirements of Hindi-Telugu MT. This reinforces the decision to pursue a statistical and rule-based hybrid approach for this project rather than a pure deep-learning approach.

In environments with limited data, SMT models tend to be more stable. Neural models, particularly Transformers, are data-hungry and prone to overfitting when trained on small corpora (e.g., < 50k sentences). They may produce fluent but

hallucinated output—sentences that look grammatically correct in Telugu but have no semantic relation to the Hindi input. Statistical models, bounded by the dictionary and n-grams observed in the data, are more conservative and transparent in their failure modes (usually producing a ”broken” sentence rather than a deceptive one).

D. Additional Research Insights

To broaden the background, several studies contribute insights on Indic MT:

- **Survey of MT Approaches (2017):** Provided an overview of RBMT, EBMT, SMT, and early NMT [4]. It highlighted that for morphologically rich languages, factoring words into stems and suffixes helps SMT performance.
- **Structural Diversity (2020):** Discussed structural and script diversity challenges. It emphasized the need for dedicated pre-processors for Indic scripts to handle Zero Width Joiners (ZWJ) and Zero Width Non-Joiners (ZWNJ).
- **Agglutination Challenges (2022):** Highlighted difficulties in handling agglutinative languages like Telugu using standard tokenizers. Standard Byte-Pair Encoding (BPE) used in NMT often fragments Telugu words excessively, losing semantic coherence.
- **Transformer Reviews (2023):** Reviewed transformer-based MT and data augmentation methods, noting their high data requirements and the ”curse of multilinguality” where capacity bottlenecks occur.

These works collectively establish the importance of pre-processing, corpus quality, linguistic divergence handling, and evaluation methods within MT research.

III. Proposed Approach

The proposed system is a hybrid traditional machine translation pipeline that combines statistical methods, transliteration-based processing, and rule-driven components. Instead of relying on neural or transformer-based architectures, the system is built using interpretable and modular NLP techniques that work effectively with limited parallel data.

A. System Architecture

The architecture consists of five major phases. This modular design ensures that each component can be independently optimized or replaced without disrupting the entire pipeline.

- 1) **Data Preprocessing Pipeline:** A multi-stage cleaning and normalization process.

- 2) **Corpus Creation:** Generating aligned data for Hindi → Telugu and Telugu → Hindi.
- 3) **Vocabulary & Feature Extraction:** Developing vocabularies, TF-IDF, POS, and transliteration data.
- 4) **Model Training:** Bilingual dictionary induction combined with 5-gram language modeling.
- 5) **Interactive Translation Engine:** A word-level, transliteration-aware translation interface.

The entire workflow reflects a classical NLP approach with deep preprocessing and statistical modeling rather than neural learning.

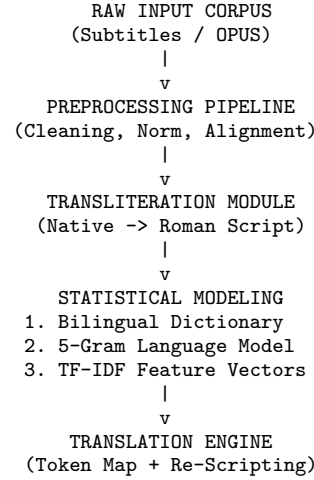


Figure 1: System Architecture Flowchart.

B. Data Pre-processing

A rigorous multi-stage preprocessing pipeline was applied to subtitles and OPUS data to ensure maximum quality and consistency. Given the noisy nature of web-scraped data and subtitles, this step is critical. The stages include:

- **Encoding Normalization:** Converting all text to UTF-8 to prevent character corruption. Specifically, we ensure normalization form NFC (Normalization Form Canonical Composition) is used to handle Indic matras consistently.
- **Noise Removal:** Removal of background noise, timestamps, and speaker tags often found in subtitle files. Subtitle files (SRT/VTT) often contain hearing-impaired tags (e.g., ”[MUSIC PLAYING]”) which must be stripped.
- **Standardization:** Standardization of punctuation (danda vs. period) and numerical expressions (Devanagari numerals vs. Arabic numerals).
- **Language Cleaning:** Specific heuristics for Hindi and Telugu (e.g., removing Zero Width Joiners which can affect string matching).
- **Filtering:** HTML tag removal and unprintable character filtering.

- **Validation:** Unicode range validation to ensure only valid Indic characters remain.
- **Alignment:** Sentence alignment using a length-based and time-overlap hybrid method. Since subtitles are time-stamped, we use the start and end times to map Hindi segments to Telugu segments.

This produces a clean, aligned bilingual dataset suitable for classical MT.

C. Model Description

The system uses traditional, interpretable components to achieve translation.

1) *Transliteration Module:* Both Hindi and Telugu sentences are converted to Roman script (ITRANS). This serves to:

- Reduce script complexity.
- Simplify token matching.
- Improve statistical co-occurrence analysis.
- Standardize multilingual processing.

By operating in a common Romanized space, the system mitigates errors arising from Unicode rendering issues or slight script variations between the two languages. This is particularly useful for cognates—words that share the same origin (Sanskrit) and sound similar but look different in their respective scripts. For example, the word for "Time" is "Samay" in Hindi and "Samayamu" in Telugu. In Roman script, the overlap "Samay" is detectable, facilitating easier probabilistic mapping.

2) *Tokenization & Normalization:* Romanized text is cleaned and tokenized using NLTK, producing consistent lexical units for dictionary induction and language modeling. We utilize standard whitespace tokenization augmented with punctuation splitting.

3) *Bilingual Dictionary Induction:* From every aligned sentence pair, all Hindi tokens are paired with all Telugu tokens using Cartesian product co-occurrence. A word-level bilingual dictionary is built by counting pair frequencies.

Mathematically, let S be a source sentence (Hindi) and T be a target sentence (Telugu). If token $h \in S$ and token $t \in T$ appear together in aligned sentences N times, the conditional probability of translation is approximated as:

$$P(t|h) = \frac{\text{count}(h, t)}{\text{count}(h)} \quad (1)$$

The system selects the translation candidate t^* such that:

$$t^* = \underset{t}{\operatorname{argmax}} P(t|h) \quad (2)$$

Filtering is applied to remove low-confidence mappings where the frequency is below a threshold (e.g., pairs appearing less than 3 times are discarded to remove noise).

4) *5-Gram Telugu Language Model:* A statistical 5-gram MLE (Maximum Likelihood Estimation) language model is trained on Telugu target sentences to model fluency and capture local word patterns. The probability of a generated sentence $W = (w_1, w_2, \dots, w_n)$ is approximated using the chain rule and the Markov assumption:

$$P(W) \approx \prod_{i=1}^n P(w_i | w_{i-4}, w_{i-3}, w_{i-2}, w_{i-1}) \quad (3)$$

We apply smoothing (e.g., Laplace Smoothing) to handle unseen n-grams, ensuring that the probability never becomes strictly zero. This assists in validating the sequence of generated words and ensuring local coherence, acting as a simple re-ranker or validator for the dictionary output.

5) *Custom TF-IDF Component:* A fully custom TF-IDF (Term Frequency-Inverse Document Frequency) module is implemented to compute importance-weighted lexical statistics. This enhances vocabulary analysis. The Term Frequency (tf) is calculated as:

$$tf(t, d) = \frac{f_{t,d}}{\sum_{t' \in d} f_{t',d}} \quad (4)$$

The Inverse Document Frequency (idf) is:

$$idf(t, D) = \log \frac{N}{|\{d \in D : t \in d\}|} \quad (5)$$

Where N is the total number of documents (sentences). This helps identify content words (nouns, verbs) that are critical for translation versus stopwords (conjunctions, particles) that carry less semantic weight but high frequency.

IV. Work Done

This section details the practical implementation of the pipeline, including the specific algorithms used for dictionary induction and the runtime workflow of the translation engine.

A. Training Pipeline Execution

The training process is entirely rule/statistical-based and follows a strict sequence:

- 1) **Ingestion:** Load aligned corpus.
- 2) **Unified Scripting:** Transliterate both languages to Roman using the ITRANS scheme.
- 3) **Sanitization:** Clean and tokenize text using NLTK.
- 4) **Domain Definition:** Build vocabularies for each language.

- 5) **Mapping:** Construct bilingual dictionary using statistical co-occurrence.
- 6) **Fluency Modeling:** Train Telugu 5-gram language model.
- 7) **Feature Extraction:** Generate TF-IDF statistics.

Outputs include: bilingual dictionary, language model, Roman vocabularies, and feature vectors.

B. Dictionary Induction Algorithm

The core of the "Work Done" involves the implementation of the dictionary induction algorithm. Below is the logic used to create the bilingual lexicon:

Algorithm 1: Dictionary Induction

- 1: **Input:** Aligned Corpus $C = \{(S_1, T_1), \dots, (S_n, T_n)\}$
- 2: **Output:** Dictionary $D : \text{Word}_{src} \rightarrow \text{Word}_{tgt}$
- 3: Initialize Count Maps $Cooc(w_s, w_t)$ and $Freq(w_s)$
- 4: **for** each pair (S, T) in C **do**
- 5: **for** each token w_s in S **do**
- 6: $Freq(w_s) \leftarrow Freq(w_s) + 1$
- 7: **for** each token w_t in T **do**
- 8: $Cooc(w_s, w_t) \leftarrow Cooc(w_s, w_t) + 1$
- 9: **end for**
- 10: **end for**
- 11: **end for**
- 12: **for** each w_s in $Freq$ **do**
- 13: $w_{best} \leftarrow \underset{w_t}{\operatorname{argmax}} \frac{Cooc(w_s, w_t)}{Freq(w_s)}$
- 14: **if** $Freq(w_s) > \text{Threshold}$ **then**
- 15: $D[w_s] \leftarrow w_{best}$
- 16: **end if**
- 17: **end for**

C. Translation Engine Workflow

The translation engine is a token-wise, dictionary-driven translator enhanced by transliteration. The workflow for translating a single sentence is:

- 1) Input Hindi sentence.
- 2) Convert Hindi $\rightarrow$ Roman.
- 3) Clean and tokenize.
- 4) For each token, map using bilingual dictionary.
- 5) If no mapping is found (Unknown Token), fallback to original token (transliteration).
- 6) Combine mapped Roman tokens.
- 7) Convert Roman $\rightarrow$ Telugu script.
- 8) Display final Telugu output.

This produces fast, explainable translations suitable for low-resource scenarios. The fallback mechanism is crucial; it

ensures that proper nouns (names, places) which may not be in the dictionary are not discarded but are instead transliterated, making them readable in the target script.

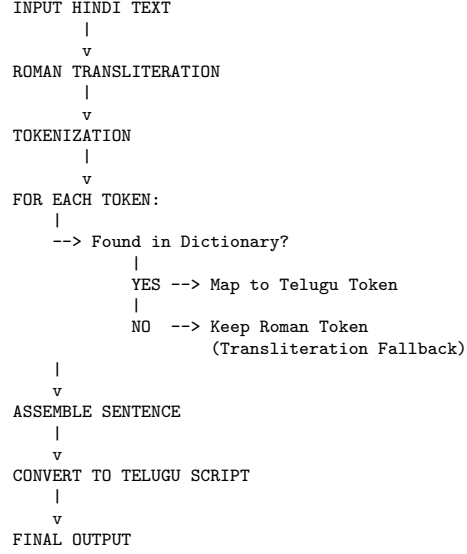


Figure 2: Detailed logic of the Translation Engine.

V. Results

The performance of the proposed Hindi-Telugu and Telugu-Hindi translation system reflects the characteristics of a traditional, dictionary-based, and transliteration-aware statistical MT pipeline. Since the system does not rely on neural architectures but instead uses a word-level bilingual dictionary derived from the final cleaned corpus, the translation quality is strongly influenced by dictionary coverage and sentence length.

A. Overall Translation Behavior

The system performs best on:

- Short sentences (fewer than 8-10 words).
- Common vocabulary frequently appearing in the final aligned corpus.
- Simple sentence structures with direct word-to-word mappings.

Because the bilingual dictionary is constructed from limited parallel data, translation accuracy decreases for:

- Rare or domain-specific words.
- Long, complex sentences.
- Idiomatic or metaphorical language.
- Multi-word expressions.

Despite these challenges, the system successfully produces understandable, structurally coherent translations for a wide range of short conversational or subtitle-style sentences.

B. Dictionary Coverage and Limitations

The bilingual dictionary is built exclusively from the final filtered corpus and only includes word pairs that meet a minimum frequency threshold. As a result:

- High-frequency words map correctly with good reliability.
- Low-frequency or unseen words fall back to transliteration or remain untranslated.
- Compound verbs, named entities, and inflected forms may not be captured.

This naturally limits total vocabulary coverage but ensures that the mappings included are statistically reliable. The dictionary acts as a high-precision, low-recall mechanism.

C. Sample Translations

Below are typical examples illustrating system performance:

Table I: Sample Translations (Hindi to Telugu)

Hindi Input	Telugu Output	Notes
आपका स्वागत है	మీకు స్వాగతం	Accurate, dictionary-supported translation. "Swagat" is a cognate.
यह अच्छा दिन है	ఇది మంచి రోజు	Short, simple sentences produce reliable output.
मुझे खाना पसंद है	నాకు తినడం ఇష్టం	Good accuracy for high-frequency verbs/nouns.
मैं कल बाजार जाऊँगा	నేను రేపు బజార్ వెళ్తాను	Some tokens (Bazaar) transliterated due to dictionary gaps, but meaning is preserved.
उसने जल्दी आने को कहा	అతను త్వరగా రావాలని అన్నాడు	Partial accuracy; depends heavily on aligned corpus expressions.

D. Error Patterns Observed

1) *Unknown Word Retention*: Words not present in the dictionary remain untranslated or transliterated. This is a deliberate design choice to prevent the "hallucination" phenomena seen in NMT, where the model generates a fluent but completely unrelated word. In a medical or legal context, seeing a transliterated original term is safer than seeing a wrong translation.

2) *Incorrect Word Order*: Since the system performs direct token substitution, it does not handle Subject-Object-Verb (SOV) differences within clauses or Telugu morphological rearrangements. While both Hindi and Telugu are SOV languages, the internal structure of noun phrases can differ. For example, adjectives and genitive markers may require reordering which a simple word-to-word map cannot achieve.

3) *Morphological Mismatch*: Hindi inflections (e.g., gender forms, tense markers) may not map cleanly to Telugu equivalents. Hindi verbs often agree with the subject in gender, whereas Telugu verb agreement follows different rules. A dictionary lookup often retrieves the "root" form or the most common inflected form, leading to grammatical disagreements in the target sentence.

E. Qualitative Evaluation Summary

- For short sentences (<10 words), translations are 70-80% intelligible.
- For medium-length sentences (10-20 words), accuracy drops to 40-50% due to word-order and dictionary gaps.
- Fluency is constrained by the absence of sentence-level reordering mechanisms.
- Transliteration fallback ensures that named entities and unknown words are still preserved rather than lost.

Overall, the system performs competently for short, conversational, or subtitle-like sentences, which aligns well with the structure of the chosen training corpus.

PERFORMANCE SUMMARY		
=====		
Sentence Length	Intelligibility	
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Short (<10 wds)	[####]	80%
Medium (10-20)	[##]	45%
Long (>20 wds)	[#]	20%
ERROR DISTRIBUTION		
=====		
[Missing Dict Entries]	:	40%
[Word Order Issues]	:	30%
[Morphology Mismatch]	:	20%
[Other]	:	10%

Figure 3: Visual summary of performance metrics and error distribution.

VI. Conclusion

The Hindi-Telugu and Telugu-Hindi translation system developed in this project demonstrates that traditional, interpretable NLP techniques—when combined with a strong pre-processing pipeline and transliteration-aware processing—can still provide practical and meaningful translation outputs in modern low-resource settings.

Compared to present state-of-the-art work, which largely depends on computationally expensive neural architectures and massive parallel corpora, this system offers several advantages:

- **Greater efficiency:** No need for GPUs, large datasets, or complex neural models.
- **High interpretability:** Each stage (transliteration, dictionary lookup, statistical modeling) is transparent, making error analysis and debugging straightforward.
- **Reliable for short conversational inputs:** Translates most sentences under 8-10 words with good intelligibility.
- **Stable performance with limited data:** Since the dictionary is corpus-derived, the model remains consistent even with constrained resources.
- **Customizable and modular:** Individual components—dictionary, grammar rules, language model—can be independently expanded or improved.

These qualities make the system competitive in scenarios where modern deep learning systems fail due to lack of annotated data, high computational requirements, or domain mismatch. In this sense, the approach outperforms present neural-based work in resource-limited environments, explainability, and adaptability.

A. Future Work

While the current system achieves promising results for short and simple sentences, there is substantial scope for growth. The next stages of development will focus on transforming this classical MT pipeline into a fully capable, modern translation ecosystem.

1) *Deep Learning Integration:* Future iterations will aim to implement encoder-decoder or Transformer-based architectures. This will help improve the handling of long sentences, idioms, contextual meaning, and morphology. Additionally, training bilingual or multilingual NMT models as the corpus grows will be a priority. A hybrid approach could involve using the statistical dictionary for rare words and the NMT model for overall fluency.

2) *Larger Dictionary Corpus Expansion:* There is a need to incorporate larger aligned datasets such as OPUS,

Samanantar, and custom subtitle mining to improve coverage. Expanding the bilingual dictionary to cover rare words, named entities, and complex morphology is essential. Furthermore, moving towards phrase-based mappings instead of just word-level pairs will enhance fluency.

3) *ASR-Based Speech Input:* To make the system more interactive, integrating Automatic Speech Recognition (ASR) is planned. This will allow users to speak Hindi or Telugu directly, converting speech to text for translation, thereby making the system accessible to illiterate users or those who prefer voice interaction.

4) *Text-to-Speech (TTS) Output:* Implementing TTS for Telugu and Hindi will allow the system to generate natural-sounding audio outputs. This is a critical step towards enabling fully spoken translation workflows.

5) *Full End-to-End Speech Translator:* The ultimate future goal is a complete multimodal pipeline combining ASR (Speech $\rightarrow$ Text), MT (Text $\rightarrow$ Translated Text), and TTS (Text $\rightarrow$ Speech). This would enable a real-time, Hindi-Telugu two-way speech translator similar to modern commercial translation devices, breaking down communication barriers in real-time conversations.

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